



Test Plan

**Online Fact-Checking System
(Sebenarnya.my)**

Version: 1.0.0

Date: 4 January 2026

Project Title: SeBenarnya.my Independent Verification and Validation		SeBenarnya.my SEBENARNYA.MY 
Date: 4 January 2026	Test Plan ID: SeBenarnya_TP_1.0.0	

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1.0 INTRODUCTION

1.1 Purpose

This Test Plan defines the testing methodology, scope, resources, and responsibilities required to verify the functionalities of the Sebenarnya.my Verification System.

Objectives include:

1. Ensuring the system meets SRS requirements.
2. Validating multi-language processing (Malay, English, Mandarin).
3. Testing AI-based classification accuracy & reliability.
4. Ensuring correct functionality for Fact-Checkers, Admins, and Users.
5. Ensuring system stability, correctness, and secure operations.
6. Validating AI behaviour quality, explainable AI relevance, adversarial input handling, and graceful handling of third-party AI failures.

1.2 Scope

The scope of testing covers black-box functional testing for all core features developed in the Sebenarnya.my Verification System, including multi-language detection, language-specific routing, AI classification, cross-language duplicate detection, translation validation, and multi-language reporting. This Test Plan includes system testing, integration testing, regression testing, confirmation testing, acceptance testing, and security testing, excluding unit testing and performance/load testing or any test types not mentioned in this document. The testing activities will focus on verifying that the system meets all stated requirements in the SRS and ensuring that the system operates accurately, reliably, and is fit for its intended purpose. For AI-based features, testing includes validation of explainable AI consistency, confidence handling, adversarial and ambiguous input behaviour, and AI service failure scenarios in addition to functional output verification.

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1.3 References

This document is prepared in reference to the following documents:

1. ATMS Test Plan. (2013, April 16). Malaysian Software Testing Board.
2. Glossary of banking terms and phrases. (2020, September 12). HelpWithMyBank.gov.
<https://www.helpwithmybank.gov/glossary/index-glossary.html>.
3. GeeksforGeeks. (2022, June 28). *Software testing | Basics*.
<https://www.geeksforgeeks.org/software-testing-basics/>
4. IBM. (n.d.). *What is software testing and how does it work?* IBM - United States.
<https://www.ibm.com/topics/software-testing>
5. JavaTpoint. (n.d.). *Software testing tutorial*. www.javatpoint.com.
<https://www.javatpoint.com/software-testing-tutorial>
6. *Test Plan Tutorial: A Guide To Write A Software Test Plan Document From Scratch*. (n.d.).
<https://www.softwaretestinghelp.com/how-to-write-test-plan-document-software-testing-training-day3/>
7. Thomas Hamilton. (2022, October 29). *What is software testing? Definition*. Guru99.
<https://www.guru99.com/software-testing-introduction-importance.html>
8. Thomas Hamilton. (2022, September 17). *TEST PLAN: What is, how to create (with example)*. Guru99.
<https://www.guru99.com/what-everybody-ought-to-know-about-test-planing.html>

The following documents provide the test basis for the test design:

1. SeBenarnya.my Software Requirement Specifications (SeBenarnya.my_SRS)

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2.0 DETAILS OF TEST PLAN

2.1 Test Items

The following documents will provide the basis for defining the correct operation:

1. SeBenarnya.my Software Requirement Specifications (SeBenarnya.my_SRS)
2. SeBenarnya.my Software Design Specifications (SeBenarnya.my_SDS)

The test items include all system features in the SeBenarnya.my Verification System, as described in Section 2.2 below.

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2.2 Test Traceability Matrix

Function ID	Risk Level	Description	Source
F001	High	Multi-language detection (Malay/English/Mandarin)	SRS
F002	High	Route content to correct classifier	SRS
F003	High	Detect cross-language duplicates	SRS
F004	High	Validate translations before analysis	SRS
F005	High	AI classification (True/False/etc.)	SRS
F006	High	Multi-language reporting	SRS
F007	Medium	User submission	SRS
F008	Medium	Fact-checker verification workflow	SRS

2.3 Features To Be Tested

Feature ID	Name	Risk
F001	Multi-language detection	High
F002	Language routing	High
F003	Cross-language duplicate detection	High
F004	Translation validation	High
F005	AI classification	High

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F006	Multi-language reporting	High
F007	User submission	Medium
F008	Fact-checker verification workflow	Medium

2.4 Features Not To Be Tested

Aside from the features listed in Section 2.2, other aspects of the system are not covered in this test plan. The aspects include:

i. Operation Procedure

This test plan focuses on system-level testing; therefore, operational procedures are not included.

ii. Performance

This test plan only covers functional and security testing; performance testing is not included.

iii. Server Load

This test plan only covers functional and security testing; load testing is not included.

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2.5 Approach

The test in this test plan is focused on system-level testing, which includes both functional testing and security testing. Testing will be conducted using a risk-based strategy to prioritize high-risk features such as multi-language detection, AI classification, and duplicate content analysis.

This test will utilize black-box testing techniques to derive the test cases. The black-box techniques used in this test include the following:

- i. Equivalence Partitioning
- ii. Boundary Value Analysis
- iii. Decision Table Testing
- iv. State Transition Testing
- v. Use Case Testing

Testing will be carried out in two iterations. In the first iteration, core processes such as user submission, multi-language detection, and routing to the appropriate classifier will be tested. Functionality tests of other modules—such as AI classification, duplicate detection, translation validation, and reporting—will be conducted in the second iteration as outlined in Section 2.3 (Features To Be Tested). All testing activities will be executed in a dedicated test environment as specified in Section 3.2. Although black-box techniques are used, AI-based features are tested through manual execution by human testers focusing on AI behaviour validation rather than deterministic outputs. This includes verifying explainable AI relevance, confidence visibility rules, adversarial resistance, consistency of classification behaviour, and graceful handling of third-party AI service failures.

Any defects identified during testing will be addressed and fixed in the subsequent release version.

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2.6 Pass/Fail Criteria

The system must fulfil the following criteria to pass:

- I. All test cases must be passed.
- II. 0% calculation or classification error within all tested functions.
- III. No deadlock, system freeze, or execution halt during any operation.

In addition to the above requirements, all test items must also fulfil the functional and non-functional requirements stated in the SeBenarnya.my_SRS.

2.7 Suspension & Resumption Requirements

Suspension Criteria:

Testing activities will be suspended if any of the following conditions occur:

- i. The AI classification service or multi-language detection service becomes unavailable.
- ii. The system backend (API or database) cannot be accessed.
- iii. Critical modules such as translation validation or duplicate detection fail to initialize.
- iv. The testing environment is unstable or cannot load required components.

Resumption Requirements:

Testing will resume once the following conditions are met:

- i. All required system services (AI module, language detection engine, backend API, and database) are fully operational.
- ii. The testing environment is restored and stable.
- iii. Any blocking defects preventing further execution have been resolved.

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2.8 Test Deliverables

The following documents will be generated by the SeBenarnya.my system test group and will be delivered throughout the system testing phase.

Test documentation:

- i. Test Plan
- ii. Test Design Specifications
- iii. Test Case Specifications
- iv. Test Logs
- v. Test Incident Report
- vi. Test Summary Report

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2.9 Entry Criteria

The items below are required before testing can begin:

- i. SeBenarnya.my_SRS Version 1.0.0
- ii. Software Under Test (SUT): Sebenarnya.my Verification System Version 1.0.0
- iii. Required test environment and test data must be prepared and accessible.
- iv. All dependent services (AI module, language detection engine, database) must be operational.

2.10 Exit Criteria

The items below are required before the testing can be concluded:

- i. All Test Deliverables listed in Section 2.8 (Test Deliverables) have been completed and delivered to the client.
- ii. Test execution has been fully completed.
- iii. Open defects at the end of the final iteration shall meet the following conditions:
 - Zero fatal defects
 - Zero major defects
 - Less than or equal to three (≤ 3) minor defects
- iv. No unresolved critical AI misclassification, explainable AI inconsistency, or unsafe adversarial classification remains at the end of testing.

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3.0 TEST MANAGEMENT

3.1 Planned Activities

No.	Task	Activities	Target Date
1	Test Planning and Control	I. Define test project scope and objectives II. Review system requirements (SeBenarnya.my_SRS) III. Identify risks, staffing needs, and test schedule- Document the Test Plan IV. Conduct technical review of the plan V. Submission and approval of the Test Plan	25/11/2025
2	Test Analysis and Design	I. Analyse system modules and requirements II. Develop Test Design Specification III. Prepare Test Case Specification	1/12/2025
3	Test Environment	I. Set up Sebenarnya.my Verification System test environment II. Identify and acquire required test data	10/12/2025
4	Test Implementation and Execution	I. Prepare Test Procedure Specification II. Execute smoke testing III. Execute functional testing (multi-language, AI, duplicate detection, reporting) IV. Record test results V. Analyse test results	11/12/2025

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5	Documenting Test Summary Report	I. Summarise all test results and defects II. Prepare and compile Test Summary Report	31/12/2025
6	Documenting Test Closure	I. Collect test process data from completed activities II. Consolidate experience, testware, facts, and metrics III. Document Test Closure Report	4/1/2026
7	Test Complete	I. Delivery of final tested product to client	15/1/2026

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3.2 Environment & Infrastructure

3.2.1 Hardware

To be defined.

3.2.2 Software

Application Name	Sebenarnya.my Verification System
Version	1.0.0
Type of File	Web Application
Size of Application	To be defined
Size on Disk	To be defined
Operating System	Windows 11

3.2.3 Publications

The following documents are required to support system testing:

1. SeBenarnya.my Software Requirement Specifications (SeBenarnya.my_SRS) Version 1.0.0

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3.3 Responsibilities

Name	Role	Responsibilities
Meor Muhammad Anuar Firdaus bin Ainol Salleh	Test Leader	- Lead and guide the testing team efficiently - Develop the Test Plan and ensure timely delivery of the product - Assess the progress and effectiveness of test efforts - Advocate for the appropriate level of quality by ensuring critical defects are resolved
Nur Liyana Athirah binti Zailani	Test Analyst	- Identify test items to be evaluated - Define appropriate test plans and prepare necessary test data - Gather and manage test data before test execution - Analyse and evaluate the outcome of each test cycle
Nur Izzah Dania binti Hipni Shahlizal	Tester	- Perform test requirement analysis - Develop and update test cases - Prepare test data - Execute tests and log results - Analyse and report defects - Perform re-tests for fixed issues - Assist in test documentation and support activities

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3.4 Resources

3.4.1 External Resources

There are no external resources required for system testing.

3.4.2 Internal Resources

The internal Quality Assurance Team shall be responsible for reviewing all test deliverables to ensure compliance with system requirements, standards, and testing best practices.

3.5 Training

All personnel involved in this testing are required to undergo training on the Sebenarnya.my Verification System according to their respective roles.

- I. Training on the overall operation flow of the Sebenarnya.my Verification System
- II. Training on the functional flow and behaviour of each system module

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3.6 Schedule, Estimates & Cost

Task	Milestone	Days	Duration
Test Planning and Control	Submission of Test Plan for sign-off	6	25/11/2025-30/11/2025
Test Analysis and Design	Submission of Test Design Specification	10	1/12/2025-10/12/2025
Test Environment	Setup of test environment	3	11/12/2025-13/12/2025
Test Implementation and Execution	Execution of system testing for Sebenarnya.my Verification System	15	14/12/2025-28/12/2025
Documenting Test Summary Report	Submission of Test Summary	3	29/12/2025-31/12/2025
Documenting Test Closure	Submission of Test Closure	1	4/1/2026
Test Complete	Delivery of product to client	1	15/1/2026

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3.7 Risk & Contingency

Rating for Likelihood for Each Risk

Rating	Description
1	Rated as Low
2	Rated as Medium
3	Rated as High
4	Rated as Extreme

Table 3.7.1 Likelihood Rating Table

Rating for Impact for Each Risk

Rating	Description
1	Rated as Low
2	Rated as Medium
3	Rated as High
4	Rated as Extreme

Table 3.7.2 Priority Rating Table

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Function ID	Risk Description	Likelihood	Impact	Risk Exposure	Risk Treatment Plan	People Responsible	Cost
F001	Multi-language detection fails to identify the correct language (Malay/English/Mandarin)	3	3	9	Perform functional tests using cross-language test data and multiple test techniques	Tester	—

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F002	System fails to route detected language to appropriate classifier	3	3	9	Conduct black-box and state-transition testing; verify routing with multilingual inputs	Tester	–
F003	Cross-language duplicate detection fails to detect similar meaning content	4	4	16	Use semantic similarity tests and multiple test scenarios	Tester	–
F004	Translation validation produces inaccurate meaning	3	3	9	Perform accuracy checks; validate using bilingual test cases	Tester	–

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F005	AI classification misclassifies True/False/Misleading/Unverified news including adversarial inputs, explainable AI validation, and failure handling scenarios.	4	4	16	Test with balanced datasets; include regression testing after model updates	Test Analyst	–
F006	Multi-language reporting produces inaccurate or incomplete output	3	2	6	Perform functional testing and verify UI-language mapping	Tester	–
F007	User submissions are not logged correctly	3	3	9	Perform functional and data validation tests on	Test Leader	–

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					submission logging and counters		
F008	Fact-checker actions fail to update user reputation or warnings	4	3	12	Test state transitions for warning, suspension, and reputation updates	Test Leader	—

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4.0 GENERAL

4.1 Metrics

These metrics will be recorded and used to monitor project progress.

Metric List	Metric Description	Goals
Percentage of test cases passed vs total test cases executed	Metric to monitor the number of test cases that pass during execution	All high-risk test cases passed
Percentage of baseline requirements covered vs total test cases	Metric to monitor coverage for all baseline requirements	100%
Percentage of test cases executed	Metric to monitor the total execution progress of all planned test cases	100%
Project duration vs plan (variance)	Metric to monitor whether the project timeline meets the planned schedule	Allow 5% delay

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4.2 Glossary

Term	Definition
SRS	Software Requirement Specification
SDS	Software Design Specification
TP	Test Plan
AI	Artificial Intelligence used for classification
NLP	Natural Language Processing module for multilingual detection

4.3 Incident Classification

Impact

- Mission Critical:**
Sebenarnya.my Verification System fails completely, or the application becomes unusable.
- Major:**
Severe problems occur but are still fixable; major functions may be degraded.

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- **Minor:**

Issues that do not affect core functionality or usability; minor deviations from requirement or design specifications.

Priority

- **Immediate:**

Must be fixed as soon as possible; blocking further testing or system operation.

- **Delayed:**

System is unstable; incident must be resolved before the next test cycle or before product release.

- **Deferred:**

Defect can be postponed due to low impact, time constraints, or cost considerations.